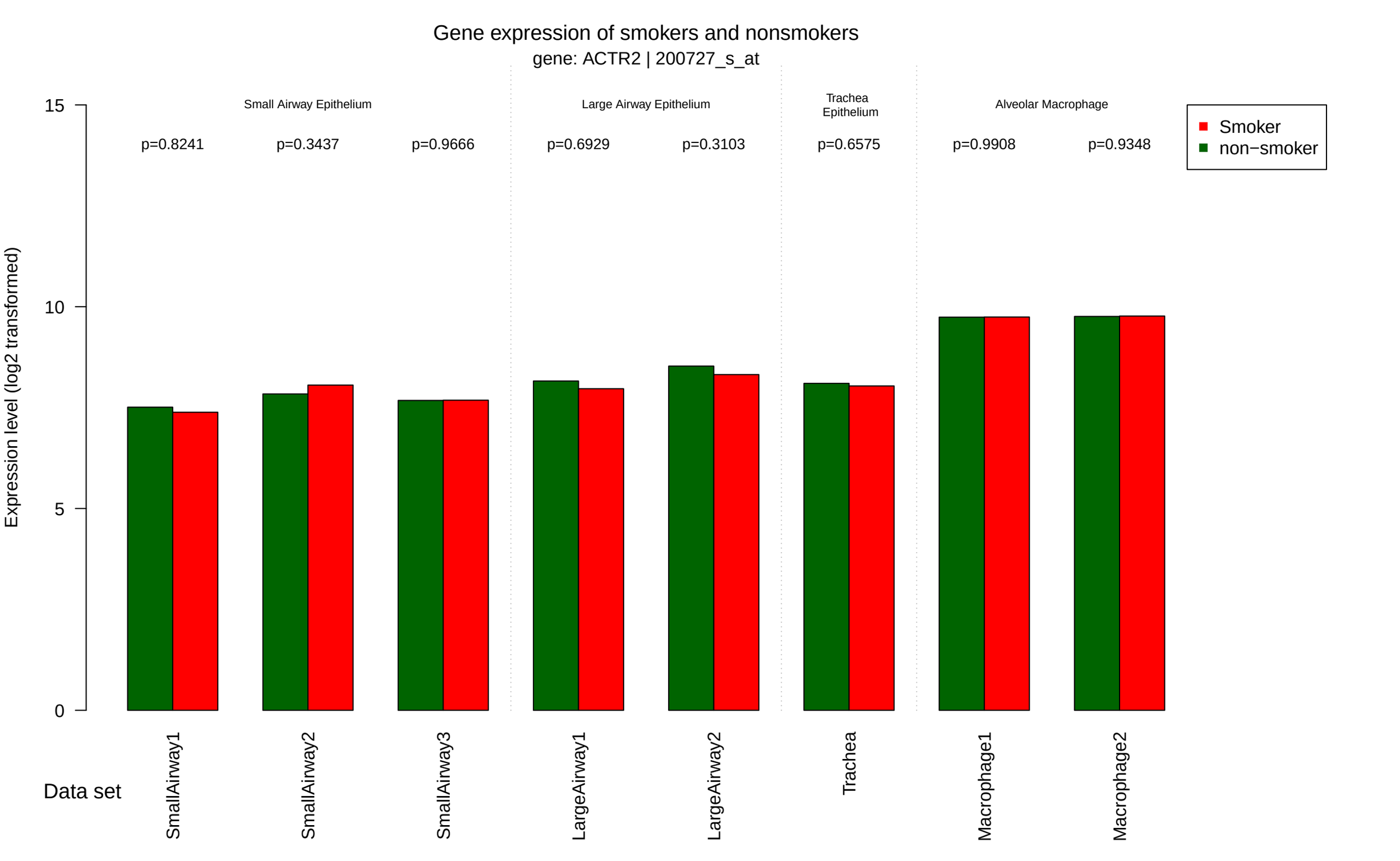
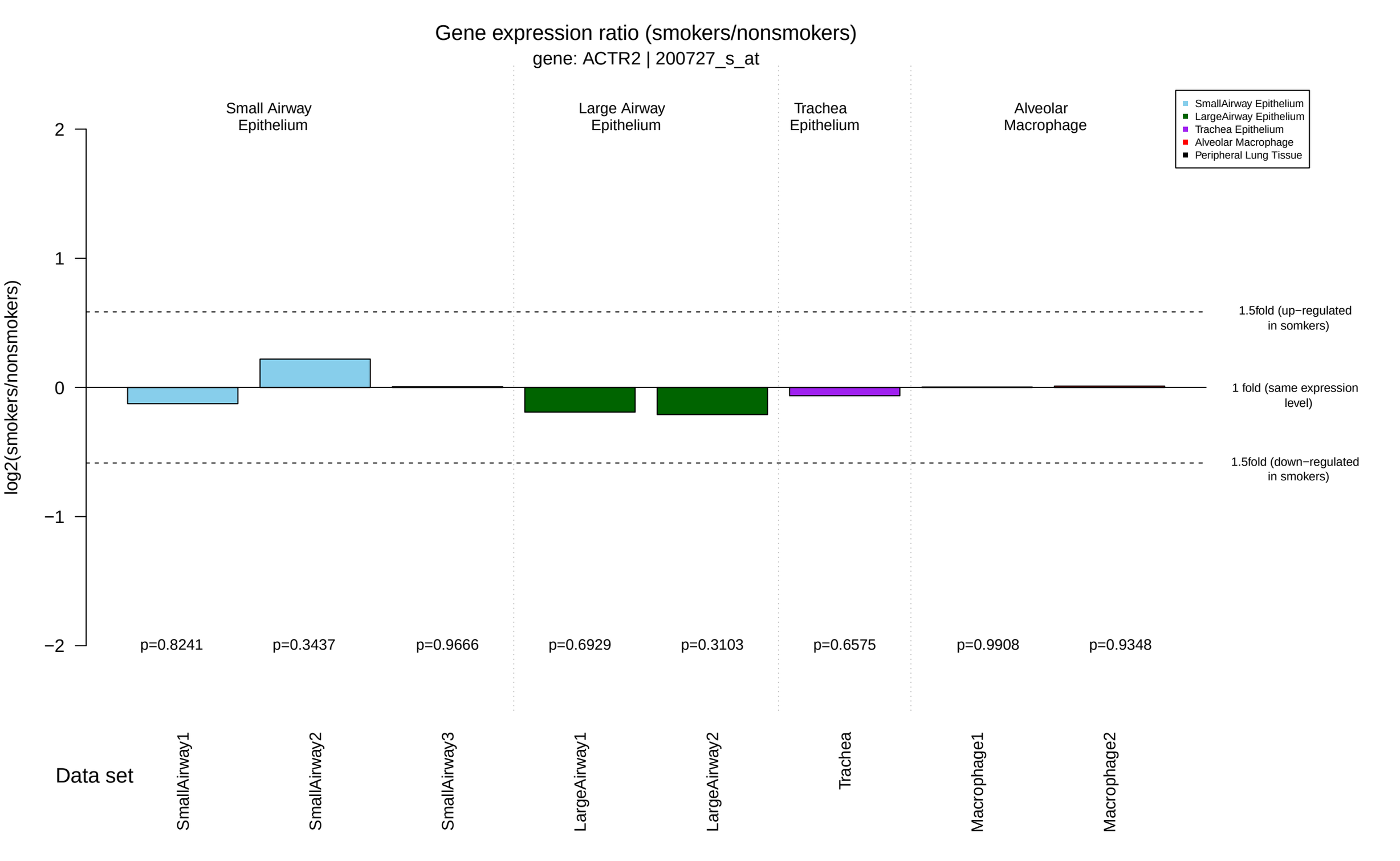
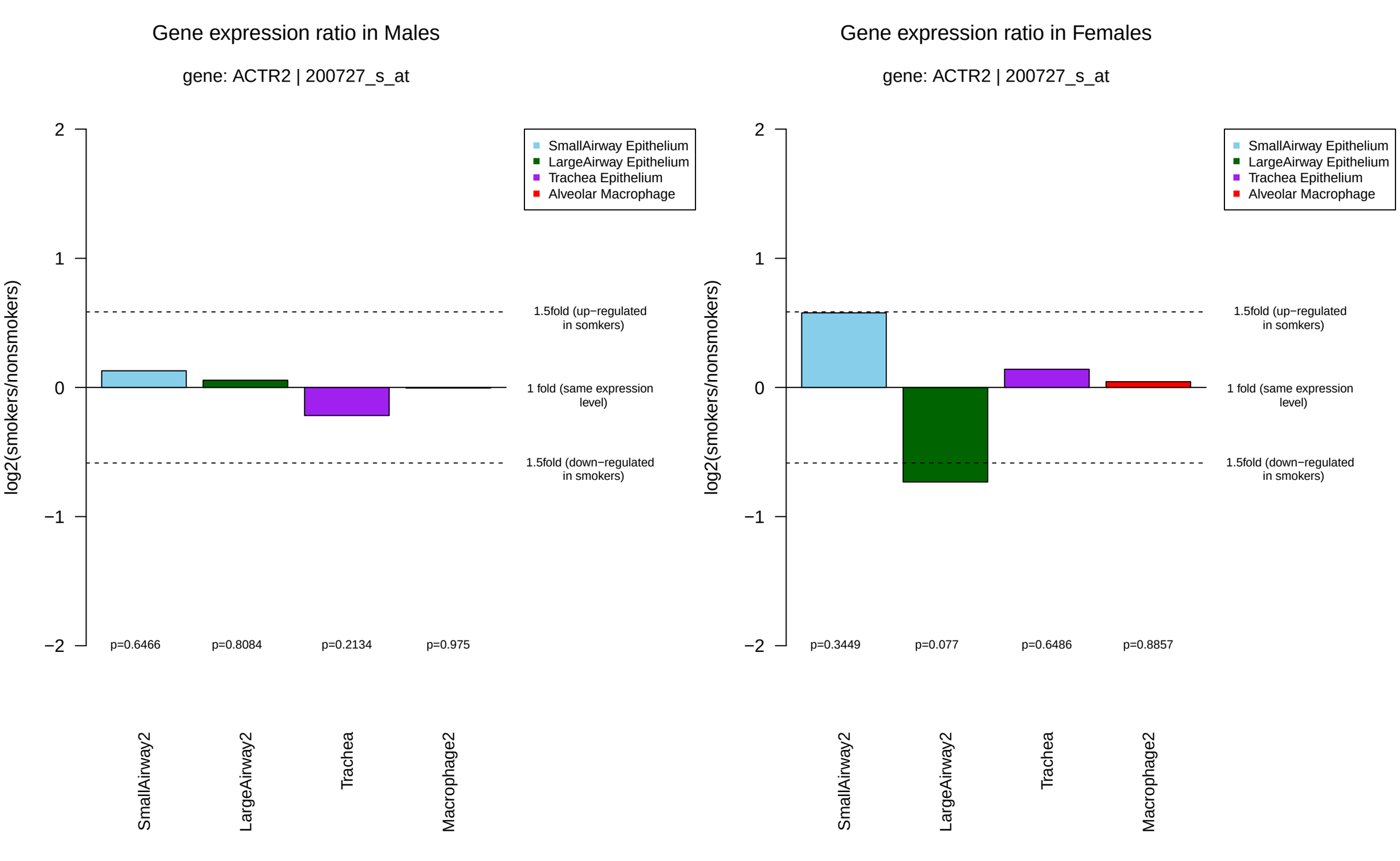
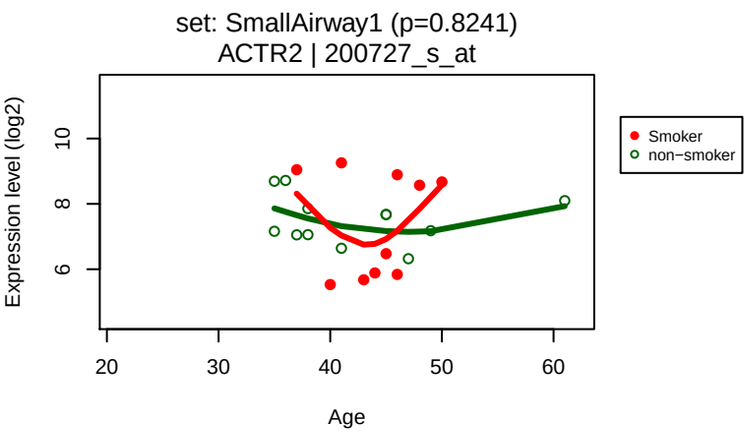
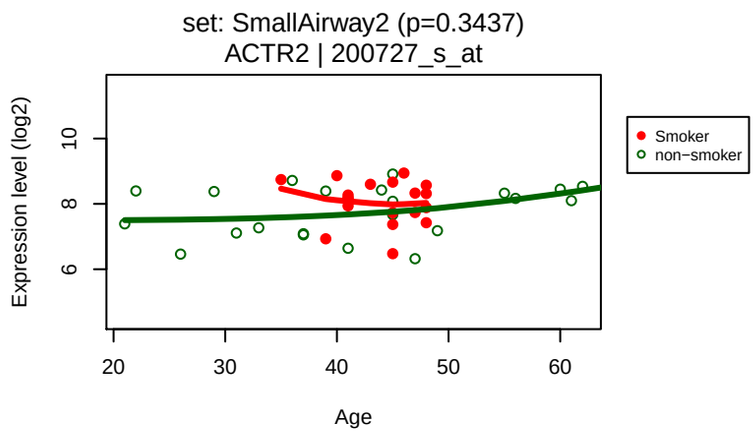
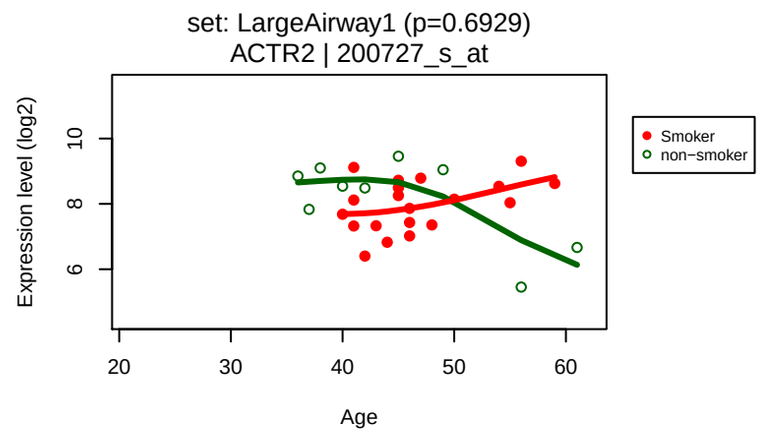

Gene expression of smokers and nonsmokers
gene: ACTR2 | 200727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8241
p=0.3437
p=0.9666
p=0.6929
p=0.3103
p=0.6575
p=0.9908
p=0.9348
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

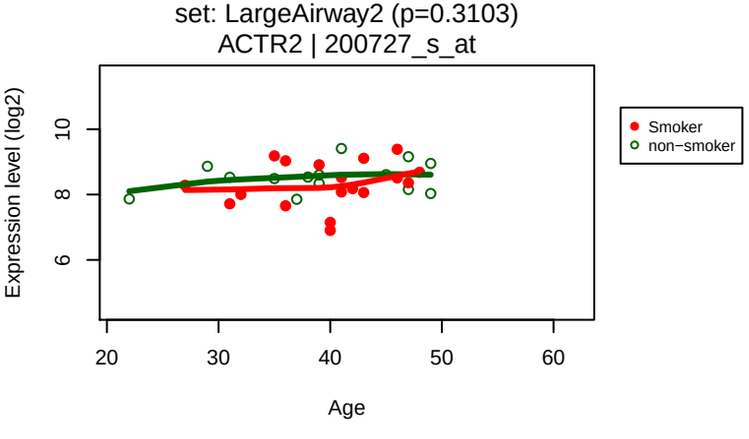

Gene expression ratio (smokers/nonsmokers)
gene: ACTR2 | 200727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8241
p=0.3437
p=0.9666
p=0.6929
p=0.3103
p=0.6575
p=0.9908
p=0.9348
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

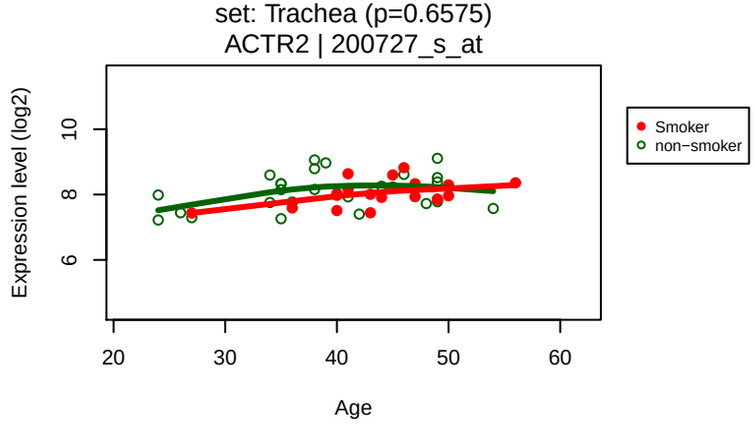

Gene expression ratio in Males
gene: ACTR2 | 200727_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6466
p=0.8084
p=0.2134
p=0.975
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR2 | 200727_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3449
p=0.077
p=0.6486
p=0.8857
SmallAirway2
LargeAirway2
Trachea
Macrophage2

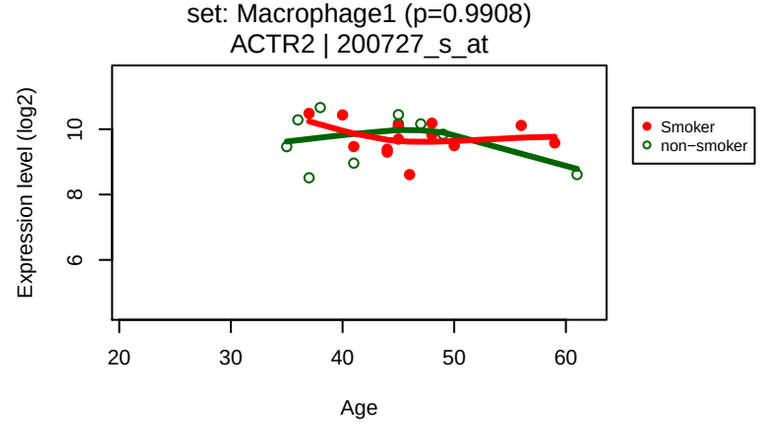

set: SmallAirway1 (p=0.8241)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker

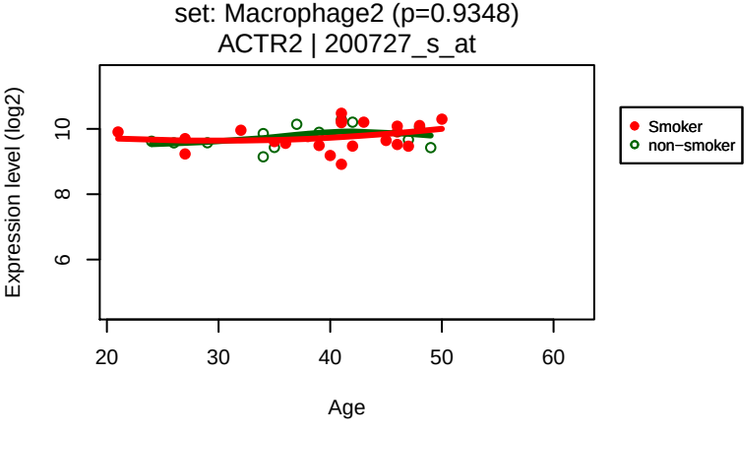

set: SmallAirway2 (p=0.3437)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker

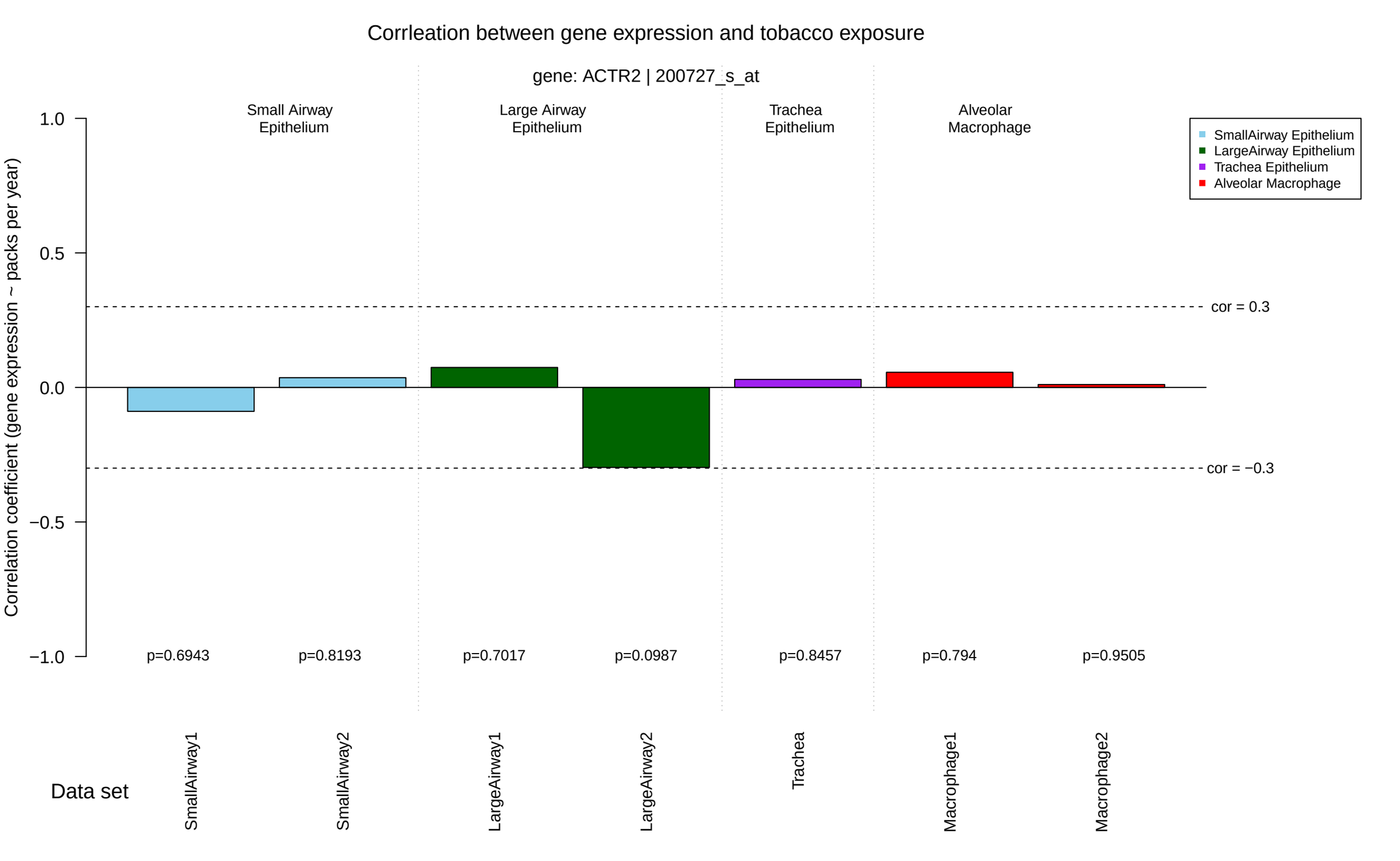

set: LargeAirway1 (p=0.6929)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3103)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6575)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9908)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9348)
ACTR2 | 200727_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR2 | 200727_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6943
p=0.8193
p=0.7017
p=0.0987
p=0.8457
p=0.794
p=0.9505
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2